EXTRACTION PROCESS

Most commercial lithium extraction is from salt-flat brines. The metal is also recovered from lithium-bearing ores

FROM HARD ROCKS

Lithium is commercially extracted primarily from five minerals: spodumene, lepidolite, petalite, amblygonite, and eucryptite

- After the ore is mined, it is crushed and roasted at 1,100°C
- It is then cooled to 65°C, milled and roasted again, this time with sulfuric acid, at 250°C, a process known as acid leaching
- During acid leaching, hydrogen in the sulfuric acid is replaced with lithium ions, to produce lithium sulfate and an insoluble residue
- Lime is then added for the removal of magnesium (an element in spodumene)
- Soda ash is used to precipitate lithium carbonate from the purified solution
- Lime slurry may also be used as a pH adjuster to neutralise excess acid from the acid leaching process

FROM BRINES

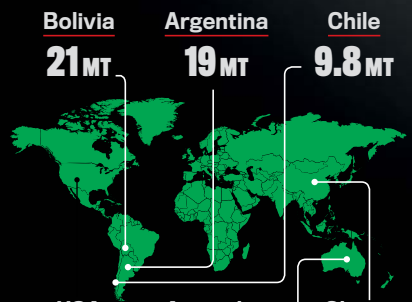
- Salt-rich water is pumped to the surface and into a series of evaporation ponds
- The water evaporates over months and a variety of salts precipitate, leaving the brine with an ever-increasing concentration of lithium
- The brine is pumped to a recovery facility for:
  - Purification
  - Chemical treatment
  - Filtration
  - Treatment with soda ash to precipitate lithium carbonate
  - Washing and drying of the lithium carbonate for the final product



WORLDWIDE DEPOSITS

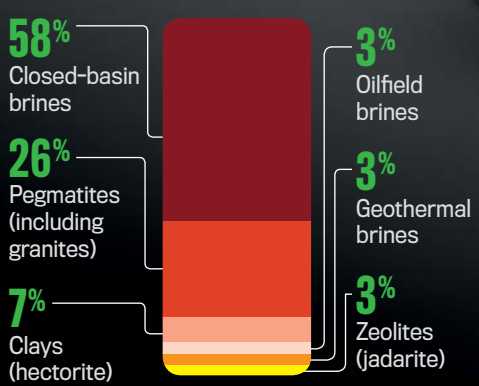
89 MT Identified

22 MT Extractible (2022)



MT: Million tonnes

MINED FROM...



Sources: USGS, STT, CSA Global, GSI & others